Yichi Zhang

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EDUCATION

Tsinghua University, *Beijing, China*
M.E. in Information and Communication Engineering, Department of Electronic Engineering *Sep. 2017 – Present.*
Advisor: Zhijian Ou, Associate Professor, Department of Electronic Engineering, Tsinghua University

Tsinghua University, *Beijing, China*
B.E. in Electronic Information Science and Technology, Department of Electronic Engineering *Sep. 2013 – Jul. 2017*
Overall GPA: 84.5/100 (3.46/4.00)
Bachelor Thesis: Application Research of Bayesian Neural Network Based on Stochastic Gradient MCMC (Chinese)
Thesis Advisor: Zhijian Ou, Associate Professor, Department of Electronic Engineering, Tsinghua University

University of California, Davis, *CA, USA*
Research Internship *Jun. 2019 – Sep. 2019*
Advisor: Zhou Yu, Assistant Professor, Computer Science and Engineering, University of California, Davis

University of Notre Dame, *IN, USA*
International Summer Undergraduate Research Experience Program *Jul. 2016 – Aug. 2016*
Advisor: Christian Poellabauer, Professor, Computer Science and Engineering, University of Notre Dame

RESEARCH INTERESTS

My research interest lies at the intersection of theoretical and applications of natural language processing, such as representation learning, language understanding and language generation, especially their applications in dialog systems. Besides deep learning, I am also interested in probabilistic machine learning and graphical models, e.g. MCMC sampling, Bayesian inference.

PUBLICATIONS & SUBMISSIONS

[1] **Yichi Zhang** and Zhijian Ou. Learning Sparse Structured Ensembles with Stochastic Gradient MCMC Sampling and Network Pruning. In IEEE 28th International Workshop on Machine Learning for Signal Processing (MLSP), pages 1–6. IEEE, 2018. Accepted as an oral paper. [link]

[2] **Yichi Zhang**, Zhijian Ou and Zhou Yu. Task-Oriented Dialog Systems that Consider Multiple Appropriate Responses under the Same Context. Accepted as a poster paper at AAAI 2020. [link]

[3] **Yichi Zhang**, Yinpei Dai, Huixin Wang and Zhijian Ou. Improved Learning of Word Embeddings with Dictionary Definitions and Semantic Injection. In submission to ICASSP 2020. [link]

[4] Qingyang Wu, **Yichi Zhang**, Yu Li and Zhou Yu. Alternating Recurrent Dialog Model with Large-Scale Pre-Trained Language Models. In submission to ICLR 2020. [link]

[5] Silin Gao, **Yichi Zhang**, Zhijian Ou and Zhou Yu. Paraphrase Augmented Task-Oriented Dialog Generation. In submission to ACL 2020.

[6] Yinpei Dai*, **Yichi Zhang*** (equal contribution), Zhijian Ou, Yanmeng Wang and Junlan Feng. Elastic-CRFs for Open-Ontology Slot Filling. In arXiv preprint arXiv:1811.01331, 2018. [link]

RESEARCH EXPERIENCES

Research Internship at Davis NLP Lab, University of California, Davis *Guide: Prof. Zhou Yu*

♦ **Diverse Policy Learning and Response Generation for Task-oriented Dialogs** *Jun. 2019 – Sep. 2019*

- Exploited the one-to-many problem in task-oriented dialogs. Proposed use in application of all valid system dialog actions as extra data to train a diversified policy to further generate appropriate and diverse responses.
- Developed an end-to-end response generation model which increases the task success rate from 68.9 (state-of-the-art at that time) to 77.9 on the challenging multi-domain dataset MultiWOZ.
- Accepted as a poster paper at AAAI 2020 [2].

♦ Paraphrase Based Dialog Data Augmentation *Jun. 2019 – Dec. 2019*

- Proposed a novel and effective data augmentation framework based on jointly seq2seq paraphrase and response generation, which is the first exploration of paraphrase based dialog data augmentation.
- Improved the state-of-the-art response generation results on two task-oriented dialog datasets.
- In submission to ACL 2020 [5].

♦ Dialog Response Generation with Pre-trained Language Models *Aug. 2019 – Oct. 2019*

- Investigated the use of pre-trained language model (GPT-2) for response generation in task-oriented dialogs.
- Our model achieves comparable results to state-of-the-art models but does not use any human annotations such as dialog state and dialog act.
- In submission to ICLR 2020 [4].

Research Assistant at SPMI Lab, Tsinghua University *Guide: Prof. Zhijian Ou*

♦ Word Representation Learning with Multiple External Semantic Resources *Jan. 2019 – Oct. 2019*

- Proposed a novel word embedding specialization method which leverages both word definitions in dictionary and semantic relations such as synonyms and antonyms.
- Our method achieves significant improvement on word similarity benchmarks and benefits downstream language understanding (dialog state tracking) tasks.
- In submission to ICASSP 2020 [3].

♦ Conversational Agent for Customer Service *Dec. 2017 – Jul. 2019*

- Developed a reliable customer service agent for the China Mobile Research Institute (CMRI). [code]
- Incorporated knowledge graph (KG) into traditional modular task-oriented dialog system for more flexible multi-turn response generation.

♦ Slot-filling for Unseen Slots and Values *Sep. 2018 – Dec. 2018*

- Proposed a CRF-based slot-filling model with a novel slot label descriptor to detect unseen slots and values.
- In arXiv preprint [6].

♦ Ensemble Learning under the Framework of Bayesian Neural Network *Mar. 2016 – Jul. 2017*

- Investigated gradient-based Monte Carlo sampling together with weight pruning as a novel ensemble learning method under the Bayesian neural network framework.
- Our method achieves better performance and lower cost on language modeling and image classification.
- Accepted as an oral paper at MLSP 2018 [1].

ACADEMIC EXPERIENCES

Invited Talk

- Introduction of advances in task-oriented dialog systems at CMRI Dialog Forum. [slide] *Sep. 2019*
- Introduction of sparse ensemble learning at AUAPAF 2018. [slide] *Oct. 2018*

Teaching Assistant

- Pattern Recognition, Tsinghua University *Spring. 2019*
- Computer Program Design, Tsinghua University *Fall. 2018*

Review Experience

- IEEE Access 2019, AUAPAF 2018

AWARDS

- Tsinghua University's Excellent Individual Scholarship (Top 10%) *Dec. 2014*
- Tsinghua University's Excellent Individual Scholarship (Top 10%) *Dec. 2016*

SKILLS

Language: English (TOEFL: 107 (R30/L29/S22/W26); GRE: V162/Q169/W3.5); Chinese (native)
Programming: Python, C/C++, MATLAB, Shell Script, MySQL
ML/NLP Libraries: PyTorch, TensorFlow, Theano, Scikit-learn, SpaCy, NLTK
Tools: Latex, SVN, Git